INCH-POUND

MIL-C-24640/16C
26 June 1995
SUPERSEDING
MIL-C-24640/16B
31 July 1989

MILITARY SPECIFICATION SHEET

CABLE, ELECTRIC, 600 VOLTS, LIGHT WEIGHT, MULTI-PAIR, AWG 18, TYPE 2XSW (INCLUDING VARIATION TYPES 2XSOW AND 2XSWA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-24640.

CONSTRUCTION
(Watertight)

FIRST - 7 strand copper conductor in accordance with ASTM B-286, tinned strands in accordance with ASTM B-33. AWG size and dimensions in accordance with Table I.

SECOND - Insulation material and basic construction requirements shall be in accordance with MIL-W-81044/12 and tested in accordance with the component wire ratings and requirements listed on this slash sheet. Quality conformance testing shall be in accordance with Table VI of MIL-C-24640.

THIRD - Two conductors, one black and one white, twisted together. The pitch shall be 1 " ± ¼" (inch).

FOURTH - Binder tape applied helically with overlap, at manufacturer's option.

FIFTH - Braided shield of # 38 AWG tin coated copper strands. Braid coverage shall be no less than 90% determined in accordance with MIL-C-27500.

SIXTH - Shield insulation of two polyester tapes or polyester tape and extruded jacket. Identification codes applied using Method 2. (NOTE: This step is not required for types 2XSW1 or 2XSWA1)

AMSC N/A FSC 6145

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SEVENTH - The required number of shielded, twisted pairs shall be cabled together. Cabling sequence shall be consecutive, starting with number 1 from center outward. Fillers may be used as necessary, to form a round cable.

EIGHTH - Binder tape applied helically with overlap.

CONSTRUCTION VARIANT FOR TYPE 2XSW

NINTH - Crosslinked polyolefin jacket. (See Table II for thickness).

CONSTRUCTION VARIANT FOR TYPE 2XSWA

NINTH - Crosslinked polyolefin jacket. (See Table III for thickness).

TENTH - Braided metal armor. Dimensions in accordance with Table III.

CONSTRUCTION VARIANT FOR TYPE 2XSOW

NINTH - Braided shield, tin coated copper strands. Braid must be constructed to meet Surface Transfer Impedance requirements as listed herein.

TENTH - Binder tape applied helically with overlap.

ELEVENTH - Crosslinked polyolefin jacket. (See Table IV for thickness).

TABLE I. Component wire details

Watertight conductor			Finished wire			
Wire size (AWG)	Stranding (no. x AWG)	Diameter nominal (inches)	Diameter (inches)	Test load life cycle accelerated aging tests (lbs. + 0.3%)	Low temp. at -40° C.	
					Mandrel diameter (inches)	Test load (lbs)
18	7 x 26	0.048	0.065 ± 0.003	0.250	1.0	1.0

TABLE II. Construction details.

PIN M24640/16	Type number	Number of pairs	Jacket thickness (inch) (minimum	Cable diameter (inch) min	Cable diameter (inch) max	Cable weight (pounds per 1000 feet) (max)
-01UN	2XSW-1	1	0.034	0.240	0.258	52.9
-02UN	2XSW-3	3	.041	.436	.470	175.0
-03UN	2XSW-7	7	.044	.573	.617	330.0

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TABLE III. Construction details - armored.

PIN M24640/16	Type number	Number of pairs	Jacket thickness (inch) (minimum	Cable diameter (inch) min	max	Cable weight (pounds per 1000 feet) (max)
-01AN	2XSWA-1	1	0.034	0.290	0.308	68.9
-02AN	2XSWA-3	3	.041	.486	.520	210.4
-03AN	2XSWA-7	7	.044	.623	.667	375.2

TABLE IV. Construction details - overall shield.

PIN M24640/16	Type number	Number of pairs	Diameter over shield (inch) (nom)	Jacket thickness (inch) (minimum	Cable diameter (inch) min	max	Cable weight (pounds per 1000 feet) (max)
-01UO	2XSOW-3	3	0.374	0.041	0.487	0.525	227.0
-02UO	2XSOW-7	7	.514	.044	.608	.656	401.0
-03UO	2XSOW-12	12	.702	.047	.802	.864	665.0
-04UO	2XSOW-19	19	.844	.050	.938	1.010	950.0
-05UO	2XSOW-30	30	1.080	.057	1.180	1.270	1442.0

REQUIREMENTS:

Qualification required.
Construction details: See tables I, II, III and IV.

Component wire:

For the component wire the following additional tests shall be applicable in accordance with MIL-W-81044:

WIRE RATINGS AND ADDITIONAL REQUIREMENTS:

Temperature rating: 135°C (275°F) maximum conductor temperature
Voltage rating: 600 volts (rms) at sea level.
Accelerated aging: Oven temperature, 250± 2°C (482± 3.6°F) for 6 hours.
Blocking: 150± 2°C (302± 3.6°F)
Color: In accordance with MIL-STD-104, class 1; white preferred
Flammability: 30 seconds (max); 3.0 inches (max); no flaming of tissue paper.
Humidity resistance: 5000 megohms for 1000 feet, minimum insulation resistance after humidity exposure.
Identification, striping or banding durability: 125 cycles (250 strokes) (min), 500 gram weight.

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WIRE RATINGS AND ADDITIONAL REQUIREMENTS (continued):

Impulse dielectric test:
Primary insulation (when tested in lieu of spark test): 6.0 kilovolts (peak) 100 percent test.
Finished wire: 8.0 kilovolts (peak), or 3 Khz spark test 5.5KV, 100 percent test.
Insulation resistance: 5000 megohms for 1000 feet (min).
Life cycle: Oven temperature, 200± 2°C (392± 3.6°F) for 120 hours.
Physical properties of primary insulation:
Tensile strength, 2500 lb/in^2 (min).
Elongation, 150 percent (min).
Shrinkage: 0.125 inch maximum at 300± 2°C (572± 3.6°F).
Spark test primary insulation: 1500 volts (rms), 60 Hz, 100 percent test.
Surface resistance: 500 megohms inches (min), initial and final readings.
Thermal shock: Oven temperature, 150± 2°C (302± 3.6°F).
Maximum change in measurement: 0.060 inch.
Wet dielectric test: 2500 volts (rms).
Wrap test: Mandrel wrap test required.

EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor Resistance - ohms/1000 feet at 25°C, maximum	7.11
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor and shield	1000
Component Shield to Component Shield	200
Component Shield to overall shield	500
Insulation resistance - megohms/1000 feet, minimum	
Conductor to conductor and shield	2500
Shield to shield (component shield only)	100
Shield to shield (component shield to overall shield) Type 2XSOW only	100
Conductor and shield continuity	No failure
Jacket flaws (type 2XSOW only)	No failure
Group A:	
Visual and dimensional	No failure
Capacitance:	
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	60
Capacitance unbalance - percent, maximum	15
Characteristic impedance - at 1 MHz, ohms	30 min, 53 max
Watertightness (see table XI of MIL-C-24640 for limits of water leakage)	No failure
Weight	No failure

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	Group B:	
	Cold bending	No failure
	Drip	No failure
*	Cross link proof test (percent maximum)	
	Jacket	50
	Tear - pounds per inch thickness, minimum	35
	Physicals (unaged)	
	Jacket	
	Tensile strength - lb/in^2, minimum	1300
	Elongation - percent, minimum	160
	Attenuation - at 3 MHz, dB/100 feet, maximum	Not applicable
	Gas flame (1 hour)	Not applicable
	Group C:	
	Physicals (aged)	
	Jacket	
	Air oven	
	Tensile strength - percent of unaged, minimum	60
	Elongation - percent of unaged, minimum	60
*	Hot oil immersion	
	Tensile strength - percent of unaged, minimum	50
*	Elongation - percent of unaged minimum	50
*	Permanence of printing (conductor) - cycles, minimum	Not Applicable
*	Permanence of printing (jacket) - cycles, minimum	125
	Armor - conformance to material, (type 2XSWA only) construction and coverage	No failure
	Shield - conformance to material, construction and coverage	No failure
	Heat distortion - percent of unaged, maximum	30
*	Cable sealant removability	No failure
	Shrinkage	No failure
	Mutual inductance - dB/1000 feet, maximum	Not applicable
*	Surface transfer impedance (type 2XSOW only)	
	Milliohms per meter, maximum	700
	EMP response, dB, minimum	60
	Pulse response time - ms/500 feet	
	Rise time, maximum	Not applicable
	Decay time, maximum	Not applicable
*	Group D	
	Flame propagation	No Failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Aging and compatibility (cable)	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18

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QUALIFICATION INSPECTION (continued):

Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	50
* Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	10
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure
Fungus Resistance	No failure
Abrasion resistance	No failure
Gas Flame (3 hour)	Not applicable
Component wire	No failure

Unit ordering length: 3 - 12 500 feet nominal.
19 & 30 250 feet nominal.

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Review Activities:
Army - AV, CR, ER, ME, AR, AL
Navy - EC, CG
DLA - IS

Preparing Activity:
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Agent:
DLA-IS

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